THE STATE UNIVERSITY OF ZANZIBAR (SUZA)

SCHOOL OF COMPUTING COMMUNICATION AND MEDIA STUDIES

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

GROUP ASSIGNMENT

COURSE CODE **INF 2202**

COURSE NAME **IS STRATEGY AND ACQUISITION**

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GROUP ASSIGNMENT:

To design and develop a Database Management System to Support Organization Decision Making.

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INTRODUCTION

1. Background

Prime Horizon Hotel is a prestigious 4-star luxury hotel located in the heart of Zanzibar, Tanzania. Established in 2015, the hotel has grown to become one of the preferred accommodation destinations for both business, travelers and tourists visiting Zanzibar.

The hotel features include:

- **150 Rooms** including standard rooms, deluxe rooms and executive suites
- **3 Restaurants** offering local and continental foods
- **2 Conference Halls** capable of hosting up to 300 guests
- **Wellness Center** with massage and gym facilities
- **A Swimming Pool** and rooftop lounge
- **24/7 Room Service**

Prime Horizon Hotel employs around 100 staff members across departments such as Front Office, Housekeeping, Food & Beverage, Maintenance, Sales & Marketing, and Administration. The hotel is renowned for delivering unique guest experiences through quality service and modern facilities, all within a prime location near business districts and tourist attractions. However, it is currently facing operational challenges due to outdated manual record-keeping methods.

2. Vision and Mission

2.1 Vision

To be the premier 5-star hotel in Zanzibar, Tanzania, recognised for excellence in hospitality, innovation, and guest satisfaction.

2.2 Mission

To consistently deliver exceptional guest experiences by providing high-quality accommodation, outstanding customer service, and modern facilities, while creating value for our employees, partners, and the community.

3. Business Processes

Prime Horizon Hotel operates through several key business processes:

3.2 Reservation and Booking Process

- Guests make room reservations via phone calls, emails, or walk-in inquiries
- Reception staff manually check room availability using physical room charts
- Guest details are recorded in paper registration books
- No online booking system is currently in place

3.3 Guest Check-in Process

- Guests arrive and present identification documents
- Staff manually search for reservation records
- Registration forms are filled out on paper
- Room keys are assigned manually
- Guests are escorted to their rooms by porters

3.4 Housekeeping Process

- Room status clean, occupied, maintenance is tracked manually
- Housekeeping staff receive paper lists of rooms to clean
- Completed rooms are reported back via walkies-talkie
- No real-time room status updates are available to reception

3.5 Food and Beverage Service

- Guests order food through room service or at restaurants
- Orders are recorded on paper order slips
- Charges are manually added to guest bills
- Restaurant reservations are managed using notebooks

3.6 Billing and Payment Process

- Guest bills are manually calculated at checkout
- Payments are recorded in physical cash books
- Invoices are handwritten or generated using basic spreadsheet software
- Credit card payments are processed manually

3.7 Reporting and Management Process

- Monthly reports are compiled manually using spreadsheets
- Data is extracted from various paper records
- Reports are often delayed and contain errors
- Management lacks real-time business intelligence

4. Existing Problems

Prime Horizon Hotel currently faces significant operational challenges due to its reliance on manual, paper-based systems. The following are the critical problems:

4.1 Double Booking Issues

- Manual room charts often have conflicts when two staff members book the same room
- Overbooking leads to guest dissatisfaction and loss of revenue

4.2 Slow Check-in and Check-out Processes

- Staff waste time searching through physical files and registers
- Guests experience long waiting times, leading to poor reviews
- During peak hours, experience high queues at the reception desk

4.3 Data Inconsistency and Errors

- Guest information is recorded multiple times across different departments
- Spelling errors and missing information are common
- Data difference between departments cause confusion

4.4 Difficulty in Tracking Guest History and Preferences

- No centralized system to store guest preferences example room type, dietary needs
- Opportunities for personalized service are missed

4.5 Inefficient Room Status Management

- Housekeeping and reception have different information about room status
- Guests are sometimes sent to occupied or dirty rooms
- Real-time room availability is not available

4.6 Delayed and Inaccurate Reporting

- Monthly reports take 2–3 weeks to compile
- Reports contain errors due to manual data entry
- Management cannot make timely strategic decisions

4.7 Revenue Leakage

- Manual billing errors result in lost revenue
- Some services are delivered but not recorded for billing
- Difficulty in tracking all revenue streams

4.8 Lack of Security and Data Backup

- Physical records can be lost, damaged, or stolen
- No backup system for guest information
- Compliance with data protection regulations is at risk

4.9 Poor Decision-Making Support

- No real-time data available for daily operational decisions
- Management operates based on guesswork rather than facts
- Strategic planning is hindered by lack of reliable data

5. Why a Database System is Needed

Prime Horizon Hotel requires a database management system (DBMS) to address the existing problems. The following justifications highlight why a database system is essential to Prime Horizon:

5.1 Automation of Manual Processes

- A database system will replace paper-based record-keeping
- Automated processes reduce human errors and save time

5.2 Centralized Data Storage

- All hotel data will be stored in one central location
- Authorized staff across departments can access the same information
- Eliminates data duplication and inconsistency

5.3 Real-Time Room Availability

- Room status updates will be immediate and accurate
- Front desk staff will know exactly which rooms are available
- Prevents double bookings and guest inconvenience

5.4 Quick Guest Check-in and Check-out

- Guest records will be retrieved in seconds
- Reduces waiting time and improves guest satisfaction
- Staff can focus on guest service instead of paperwork

5.5 Improved Revenue Management

- Accurate tracking of all revenue streams
- Prevents revenue leakage from unrecorded services
- Enables data-driven pricing strategies

5.6 Enhanced Guest Experience

- Guest preferences and history will be stored and accessible
- Personalized services can be offered to returning guests

5.7 Efficient Reporting and Decision-Making

- Reports can be generated instantly with accurate data
- Management can make informed strategic decisions
- Supports business planning and growth strategies

5.8 Data Security and Backup

- Digital records can be secured with passwords and encryption
- Regular backups prevent data loss

5.9 Competitive Advantage

- Technology-driven hotels attract more guests
- Efficiency leads to better reviews and increased occupancy
- Helps the hotel stand out in a competitive market

1. DATABASE SYSTEM REQUIREMENTS

Prime Horizon Hotel Zanzibar is a beachfront hotel in Paje, Tanzania, celebrated for its ocean views and Swahili-inspired architecture. Attracting both international tourists and local visitors, the hotel is implementing a new database management system to improve operational efficiency and support strategic decision-making.

2. Users of the System

The database system will be used by a range of stakeholders within and outside the hotel, each with distinct access levels and responsibilities, as outlined below.

2.1 System Administrator

The System Administrator is responsible for managing the overall database system. Key responsibilities include creating and managing user accounts and permissions, performing database backups and recovery, monitoring system performance and security, and troubleshooting system issues.

2.2 General Manager

The General Manager is responsible for overseeing all hotel operations. Key responsibilities include reviewing operational and financial reports, analyzing business performance, making strategic decisions based on data, and approving budgets and pricing strategies.

2.3 Front Desk Receptionists

The Front Desk Receptionists are responsible for handling guest interactions at reception. Key responsibilities include registering new guests and checking them in, managing room reservations and bookings, processing guest check-outs and payments, and updating guest information.

2,4 Housekeeping Staff

The Housekeeping Staff is responsible for maintaining room cleanliness and readiness. Key responsibilities include viewing room assignment lists and reporting room readiness after cleaning.

2.5 Restaurant and Food & Beverage Staff

The Restaurant and Food & Beverage Staff is responsible for managing dining services. Key responsibilities include recording guest food and beverage orders, adding charges to guest bills, and viewing restaurant reservations.

2.6 Finance and Accounting Staff

The Finance and Accounting Staff is responsible for managing financial operations. Key responsibilities include processing payments and invoices, tracking revenue and expenses, generating financial reports, and managing accounts receivable and payable.

2.7 Guests

Guests are the customers of the hotel. Their interactions with the system include providing personal information for registration, requesting services and amenities, and making reservations.

3. Functional Requirements

The following functional requirements define the specific capabilities the system must provide to support day-to-day hotel operations.

3.1 Guest Management

- The system must store detailed guest information, including name, contact details, identification type and number, nationality, and preferences.
- The system must allow staff to search for guests by name, phone number, email, or booking reference.
- The system must maintain a history of all guest stays, including dates, room types, and services used.
- The system must store guest preferences (e.g., room type, dietary requirements, and special requests) to support personalized service.

3.2 Room Management

- The system must store room information, including room number, room type, floor, capacity, rate per night, and amenities.
- The system must track real-time room status (Available, Occupied, Cleaning, Maintenance, or Reserved).

- The system must allow staff to view available rooms for specific dates.
- The system must prevent double booking of rooms for the same date range.

3.3 Reservation Management

- The system must allow staff to create, modify, and cancel room reservations.
- The system must store reservation details, including guest ID, room number, check-in date, check-out date, number of guests, and special requests.
- The system must automatically update room status when a reservation is made or canceled.
- The system must provide a calendar view of reservations to avoid overbooking.

3.4 Check-in and Check-out

- The system must support quick check-in by retrieving reservation details and updating room status when a room becomes occupied.
- The system must support check-out by calculating the total bill, processing payment, and updating room status to Available or Cleaning.
- The system must generate and print guest invoices at checkout.

3.5 Payment and Billing

- The system must record all payments made by guests.
- The system must generate accurate bills that include room charges, food and beverage services, laundry, and other services.
- The system must track outstanding balances and unpaid invoices.
- The system must support multiple payment types and currencies.

3.6 Service Management

- The system must allow recording of additional services requested by guests (e.g., room service, laundry, spa, airport transfer).
- The system must automatically add service charges to guest bills.
- The system must maintain a list of available services and their prices.

3.7 Reporting and Analytics

- The system must generate daily, weekly, monthly, and yearly reports on occupancy rates.
- The system must generate revenue reports by room type, service, and time period.
- The system must generate reports on staff performance and productivity.

3.8 User Management and Security

- The system must allow creation of user accounts with unique usernames and passwords.

- The system must assign role-based access permissions to different user types.
- The system must maintain audit logs of all user activities.
- The system must support password recovery and reset functionality.

4. Non-Functional Requirements

In addition to the functional capabilities above, the system must satisfy the following quality and operational standards.

4.1 Performance Requirements

- The system must respond to user queries within 3 seconds under normal load.
- The system must support concurrent access by at least 20 users simultaneously without performance degradation.
- Report generation must complete within 5 minutes for monthly reports.
- The system must be available 24 hours a day, 7 days a week.

4.2 Security Requirements

- All user passwords must be encrypted using strong hashing algorithms.
- The system must implement role-based access control to prevent unauthorized data access.
- The system must log all login attempts and data modifications for audit purposes.
- An automatic session timeout must occur after 15 minutes of inactivity.

4.3 Reliability and Availability Requirements

- The system must have a minimum up time of 99.5%.
- The system must have a recovery mechanism to restore data in the event of a failure.
- Data integrity constraints must be enforced to prevent invalid data entry.

4.4 Usability Requirements

- The system interface must be user-friendly and require minimal training for staff.
- Data entry forms must be intuitive, with clear labels and validation.
- The system must provide helpful error messages for invalid inputs.
- Search functionality must be quick, accurate, and support relevant filters.

4.5 Scalability Requirements

- The database schema must be designed to accommodate future expansion.
- The system must support integration with future online booking platforms and third-party systems.

4.6 Maintenance and Support Requirements

- The system should be easy to maintain, with a well-documented database schema.
- Regular updates and patches must be applied to ensure security.
- Technical support must be available for troubleshooting.

4.7 Backup and Recovery Requirements

- Full database backups must be performed automatically every night.
- Transaction logs must be maintained for point-in-time recovery.
- Backup files must be stored in a secure, off-site location.

1. DATABASE DESIGN

This section presents the database design for Prime Horizon Hotel Zanzibar. It identifies all entities, their attributes, and the relationships between them. The database supports core hotel operations, including guest management, room management, reservations, check-in/check-out, payments, and services.

2. List of Entities

The Prime Horizon Hotel database management system comprises eight tables, described below.

Entity	Description
Guest	Stores guest personal information
Room	Stores room details and status
Reservation	Stores booking information
CheckInOut	Stores check-in and check-out details
Payment	Stores payment transactions
Service	Stores additional services offered
Service Order	Stores services requested by guests
Staff	Stores hotel employee information

3. Entity Details and Attributes

Each entity and its corresponding attributes, data types, and constraints are detailed below.

Guest Entity

Attribute	Data Type	Constraints
GuestID	INT	PRIMARY KEY, AUTO_INCREMENT
FirstName	VARCHAR(50)	NOT NULL
LastName	VARCHAR(50)	NOT NULL
Email	VARCHAR(100)	UNIQUE
PhoneNumber	VARCHAR(20)	NOT NULL
Nationality	VARCHAR(50)	—
IDNumber	VARCHAR(50)	—
Preferences	TEXT	—
RegistrationDate	DATE	DEFAULT CURRENT_DATE

Room Entity

Attribute	Data Type	Constraints
RoomNo	INT	PRIMARY KEY
RoomType	VARCHAR(50)	NOT NULL
PricePerNight	DECIMAL(10,2)	NOT NULL
Capacity	INT	NOT NULL
Status	VARCHAR(20)	NOT NULL
FloorNo	INT	NOT NULL
Amenities	TEXT	—

Reservation Entity

Attribute	Data Type	Constraints
ReservationID	INT	PRIMARY KEY, AUTO_INCREMENT
GuestID	INT	FOREIGN KEY
RoomNo	INT	FOREIGN KEY

Attribute	Data Type	Constraints
CheckInDate	DATE	NOT NULL
CheckOutDate	DATE	NOT NULL
NumberOfGuests	INT	DEFAULT 1
ReservationDate	DATE	DEFAULT CURRENT_DATE
Status	VARCHAR(20)	NOT NULL
TotalAmount	DECIMAL(10,2)	—
PaymentStatus	VARCHAR(20)	—
SpecialRequests	TEXT	—

CheckInOut Entity

Attribute	Data Type	Constraints
CheckID	INT	PRIMARY KEY, AUTO_INCREMENT
ReservationID	INT	FOREIGN KEY
GuestID	INT	FOREIGN KEY
RoomNo	INT	FOREIGN KEY
CheckInTime	DATETIME	DEFAULT CURRENT_TIMESTAMP
CheckOutTime	DATETIME	—
TotalNights	INT	—
FinalAmount	DECIMAL(10,2)	—
StaffID	INT	FOREIGN KEY

Payment Entity

Attribute	Data Type	Constraints
PaymentID	INT	PRIMARY KEY, AUTO_INCREMENT
GuestID	INT	FOREIGN KEY
ReservationID	INT	FOREIGN KEY

Attribute	Data Type	Constraints
Amount	DECIMAL(10,2)	NOT NULL
PaymentDate	DATETIME	DEFAULT CURRENT_TIMESTAMP
PaymentMethod	VARCHAR(20)	NOT NULL
PaymentReference	VARCHAR(100)	—
Status	VARCHAR(20)	—

Service Entity

Attribute	Data Type	Constraints
ServiceID	INT	PRIMARY KEY, AUTO_INCREMENT
ServiceName	VARCHAR(100)	NOT NULL
Description	TEXT	—
Price	DECIMAL(10,2)	NOT NULL
Category	VARCHAR(50)	—

Service Order Entity

Attribute	Data Type	Constraints
ServiceOrderID	INT	PRIMARY KEY, AUTO_INCREMENT
GuestID	INT	FOREIGN KEY
ReservationID	INT	FOREIGN KEY
ServiceID	INT	FOREIGN KEY
Quantity	INT	DEFAULT 1
OrderDate	DATETIME	DEFAULT CURRENT_TIMESTAMP
TotalPrice	DECIMAL(10,2)	NOT NULL
Status	VARCHAR(20)	—

Staff Entity

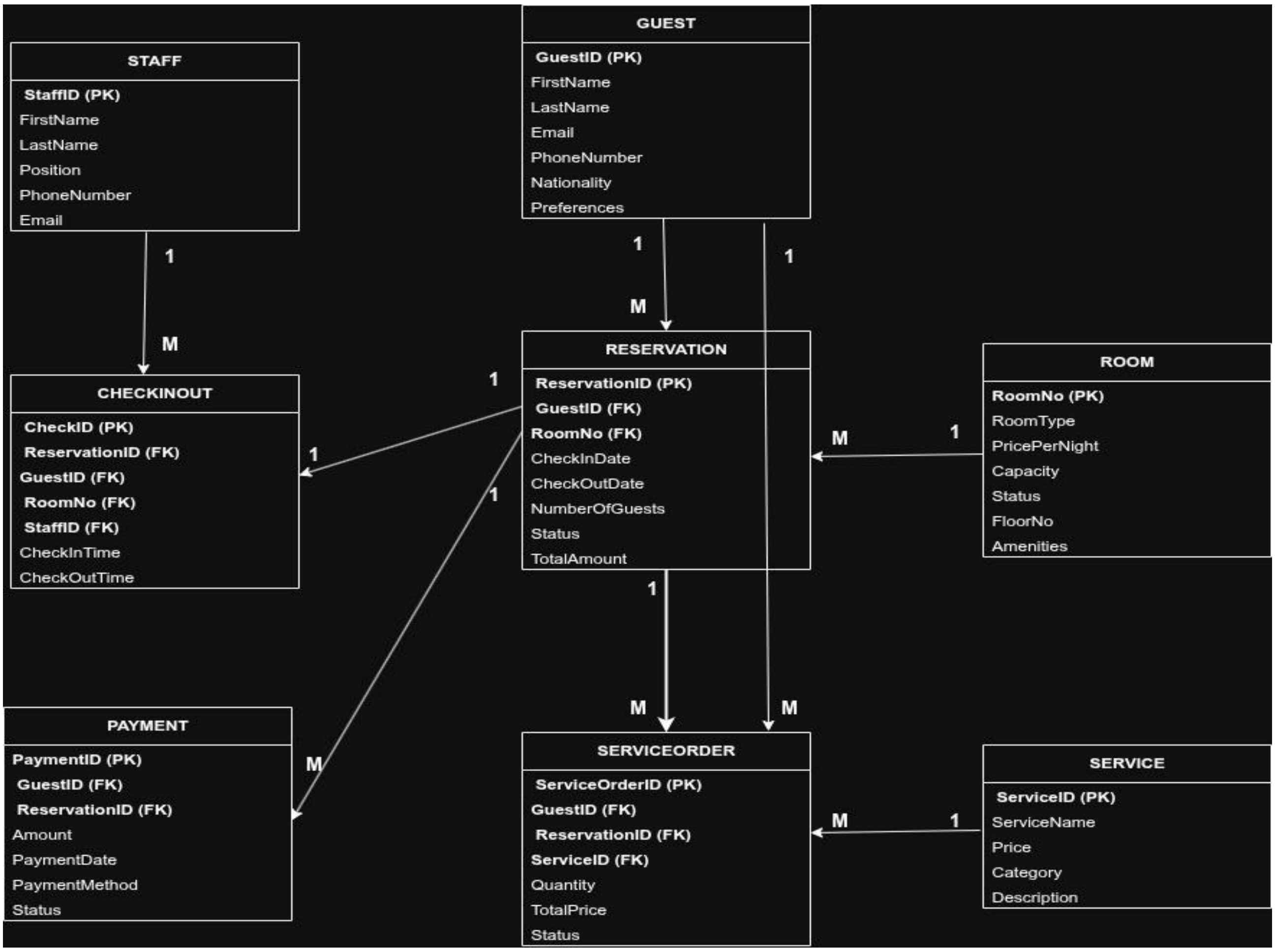
Attribute	Data Type	Constraints
StaffID	INT	PRIMARY KEY, AUTO_INCREMENT
FirstName	VARCHAR(50)	NOT NULL
LastName	VARCHAR(50)	NOT NULL
Email	VARCHAR(100)	UNIQUE
PhoneNumber	VARCHAR(20)	NOT NULL
Position	VARCHAR(50)	NOT NULL
HireDate	DATE	NOT NULL

4. Entity Relationships

The table below summarizes the relationships between entities within the database.

Relationship	Type	Description
Guest – Reservation	One-to-Many	One guest can make many reservations.
Room – Reservation	One-to-Many	One room can have many reservations over time.
Reservation – CheckInOut	One-to-One	One reservation has one check-in/check-out record.
Reservation – Payment	One-to-Many	One reservation can have many payments.
Guest – ServiceOrder	One-to-Many	One guest can order many services.
Service – ServiceOrder	One-to-Many	One service can be ordered many times.
Reservation – ServiceOrder	One-to-Many	One reservation can have many service orders.
Staff – CheckInOut	One-to-Many	One staff member can handle many check-ins/check-outs.

ER Diagram



1. DATABASE NORMALIZATION

(UNF to 3NF)

This section outlines the process of normalizing the Prime Horizon Hotel database, transitioning from Unnormalized Form (UNF) to Third Normal Form (3NF). The primary objective is to enhance database efficiency, consistency, and eliminate anomalies. In the UNF stage, all hotel data is stored in a single table that contains repeating groups and duplicate information.

Table: Hotel_Record (UNF)

ResID	GuestName	GuestPhone	GuestEmail	Nationality	RoomNo	RoomType	Price	Floor	CheckIn	CheckOut	Services	SvcPrices
1	John Smith	0712-345-678	john@email.com	USA	101	Standard	$120	1	10/07/26	15/07/26	Room Service, Laundry	$15, $20
1	John Smith	0712-345-678	john@email.com	USA	101	Standard	$120	1	10/07/26	15/07/26	Room Service, Laundry	$15, $20
2	Maria Garcia	0712-345-679	maria@email.com	Spain	202	Deluxe	$180	2	12/07/26	18/07/26	Spa, Room Service	$60, $15
3	Ahmed Hassan	0712-345-680	ahmed@email.com	Tanzania	301	Suite	$280	3	20/07/26	25/07/26	Airport Transfer	$40

Problems in UNF: the Services column holds multiple values in a single cell (a repeating group); guest data is repeated for every reservation; room data is likewise repeated for every reservation; and Price depends on Room Type rather than directly on Room No.

3. 1NF — First Normal Form

Rules applied: each cell must contain only a single, atomic value; all values within a column must share the same data type; and each row must be unique.

Converting from UNF to 1NF removed the repeating groups by creating a separate row for each service and introduced a composite primary key of ResID and ServiceName.

Table: Hotel_Record_1NF

ResID	GuestName	GuestPhone	GuestEmail	Nationality	Room No	RoomType	Price	Floor	CheckIn	CheckOut	ServiceName	SvcPrice
1	John Smith	0712-345-678	john@email.com	USA	101	Standard	$120	1	10/07/26	15/07/26	Laundry	$20
2	Maria Garcia	0712-345-679	maria@email.com	Spain	202	Deluxe	$180	2	12/07/26	18/07/26	Spa	$60
2	Maria Garcia	0712-345-679	maria@email.com	Spain	202	Deluxe	$180	2	12/07/26	18/07/26	Room Service	$15
3	Ahmed Hassan	0712-345-680	ahmed@email.com	Tanzania	301	Suite	$280	3	20/07/26	25/07/26	Airport Transfer	$40

Remaining problems in 1NF: guest information is still duplicated across rows, partial dependencies persist, and update anomalies remain possible.

4. 2NF — Second Normal Form

Rules applied: the table must already be in 1NF, and it must contain no partial dependencies.

Analysis of the 1NF table identified a partial dependency: GuestName, GuestPhone, GuestEmail, and Nationality depend only on GuestID rather than on the full composite key. The table was therefore decomposed into five separate tables, as shown below.

Table 1: Guest (2NF)

GuestID	GuestName	GuestPhone	GuestEmail	Nationality
1	John Smith	0712-345-678	john@email.com	USA
2	Maria Garcia	0712-345-679	maria@email.com	Spain
3	Ahmed Hassan	0712-345-680	ahmed@email.com	Tanzania

Table 2: Room (2NF)

RoomNo	RoomType	Price	Floor
101	Standard	$120	1

RoomNo	RoomType	Price	Floor
202	Deluxe	$180	2
301	Suite	$280	3

Table 3: Reservation (2NF)

ReservationID	GuestID	RoomNo	CheckIn	CheckOut
1	1	101	10/07/26	15/07/26
2	2	202	12/07/26	18/07/26
3	3	301	20/07/26	25/07/26

Table 4: Service (2NF)

ServiceID	ServiceName	ServicePrice
1	Room Service	$15
2	Laundry	$20
3	Spa	$60
4	Airport Transfer	$40

Table 5: Reservation_Service (2NF) — Bridge Table

ResServiceID	ReservationID	ServiceID
1	1	1
2	1	2
3	2	3
4	2	1
5	3	4

Remaining problem in 2NF: a transitive dependency remains, since Price depends on RoomType rather than directly on Room No.

5. 3NF — Third Normal Form

Rules applied: the table must already be in 2NF, and it must contain no transitive dependencies — every non-key column must depend only on the primary key.

The transitive dependency identified in 2NF — Price depending on RoomType — was resolved by moving Price into a new RoomType table.

Table 1: Guest (3NF)

GuestID	Guest Name	Guest Phone	Guest Email	Nationality
1	John Smith	0712-345-678	john@email.com	USA
2	Maria Garcia	0712-345-679	maria@email.com	Spain
3	Ahmed Hassan	0712-345-680	ahmed@email.com	Tanzania

Table 2: RoomType (3NF) — New Table

Room TypeID	Room Type	Price
1	Standard	$120
2	Deluxe	$180
3	Suite	$280

Table 3: Room (3NF)

RoomNo	RoomTypeID	Floor
101	1	1
202	2	2
301	3	3

Table 4: Reservation (3NF)

ReservationID	GuestID	RoomNo	CheckIn	CheckOut
1	1	101	10/07/26	15/07/26
2	2	202	12/07/26	18/07/26
3	3	301	20/07/26	25/07/26

Table 5: Service (3NF)

ServiceID	ServiceName	Price
1	Room Service	$15
2	Laundry	$20

ServiceID	ServiceName	Price
3	Spa	$60
4	Airport Transfer	$40

Table 6: Reservation_Service (3NF)

ResServiceID	ReservationID	ServiceID
1	1	1
2	1	2
3	2	3
4	2	1
5	3	4

1. Management Reports & Strategic Analysis

This section presents management reports generated from the hotel database and provides strategic analysis on how these reports support organizational decision-making. The reports help managers understand business performance, identify trends, and make informed strategic decisions.

2. Management Reports

Report 1: Monthly Revenue Report

Purpose: Shows hotel revenue performance by month to track financial health.

```
SELECT
    DATE_FORMAT(p.PaymentDate, '%M %Y') AS Month,
    COUNT(p.PaymentID) AS TotalTransactions,
    SUM(p.Amount) AS TotalRevenue,
    AVG(p.Amount) AS AverageTransaction,
    MIN(p.Amount) AS MinPayment,
    MAX(p.Amount) AS MaxPayment,
    COUNT(DISTINCT p.GuestID) AS UniqueGuests
FROM Payment p
WHERE p.Status = 'Success'
GROUP BY DATE_FORMAT(p.PaymentDate, '%Y-%m'),
DATE_FORMAT(p.PaymentDate, '%M %Y')
ORDER BY p.PaymentDate DESC;
```

Sample Output:

Month	TotalTransactions	TotalRevenue	AverageTransaction	UniqueGuests
July 2026	8	$4,890.00	$611.25	12
August 2026	2	$1,660.00	$830.00	4

Report 2: Room Occupancy Report

Purpose: Shows room occupancy rates by room type to identify demand patterns.

```
Report 2: Room Occupancy Report
SELECT
    rt.RoomType,
    COUNT(r.RoomNo) AS TotalRooms,
    SUM(CASE WHEN r.Status = 'Occupied' THEN 1 ELSE 0 END) AS
OccupiedRooms,
    SUM(CASE WHEN r.Status = 'Available' THEN 1 ELSE 0 END)
AS AvailableRooms,
    SUM(CASE WHEN r.Status = 'Reserved' THEN 1 ELSE 0 END) AS
ReservedRooms,
    SUM(CASE WHEN r.Status = 'Cleaning' THEN 1 ELSE 0 END) AS
CleaningRooms,
    SUM(CASE WHEN r.Status = 'Maintenance' THEN 1 ELSE 0 END)
AS MaintenanceRooms,
    ROUND((SUM(CASE WHEN r.Status IN ('Occupied', 'Reserved')
THEN 1 ELSE 0 END)
        / COUNT(r.RoomNo)) * 100, 1) AS OccupancyRate,
    rt.Price AS RoomPrice
FROM Room r
JOIN RoomType rt ON r.RoomTypeID = rt.RoomTypeID
GROUP BY rt.RoomTypeID
ORDER BY OccupancyRate DESC;
```

Sample Output:

RoomType	TotalRooms	Occupied	Available	Reserved	OccupancyRate	RoomPrice
Executive Suite	4	1	2	1	50.0%	$280
Deluxe	5	2	2	1	60.0%	$180
Standard	7	2	4	0	28.6%	$120
Presidential Suite	2	0	2	0	0.0%	$500
Family Room	3	1	2	0	33.3%	$220

Report 3: Most Popular Rooms Report

Purpose: Identifies which room types are most booked to guide inventory decisions.

```
Report 3: Most Popular Rooms by Booking Frequency
SELECT
    rt.RoomType,
    COUNT(r.ReservationID) AS TotalBookings,
    SUM(r.TotalAmount) AS TotalRevenue,
    AVG(r.TotalAmount) AS AverageBookingValue,
    COUNT(DISTINCT r.GuestID) AS UniqueGuests
FROM Reservation r
JOIN Room rm ON r.RoomNo = rm.RoomNo
JOIN RoomType rt ON rm.RoomTypeID = rt.RoomTypeID
WHERE r.Status NOT IN ('Cancelled')
GROUP BY rt.RoomType
ORDER BY TotalBookings DESC;
```

Sample Output:

RoomType	TotalBookings	TotalRevenue	AvgBookingValue	UniqueGuests
Deluxe	4	$3,960.00	$990.00	4
Standard	3	$1,320.00	$440.00	3

RoomType	TotalBookings	TotalRevenue	AvgBookingValue	UniqueGuests
Executive Suite	3	$4,200.00	$1,400.00	3
Family Room	2	$1,980.00	$990.00	2
Presidential Suite	1	$2,500.00	$2,500.00	1

Report 4: Guest Demographics Report

Purpose: Analyzes guest nationalities and preferences for targeted marketing.

```
Report 4: Guest Demographics by Nationality
SELECT
    g.Nationality,
    COUNT(g.GuestID) AS TotalGuests,
    COUNT(r.ReservationID) AS TotalBookings,
    SUM(r.TotalAmount) AS TotalSpent,
    AVG(r.TotalAmount) AS AverageSpent,
    GROUP_CONCAT(DISTINCT g.Preferences SEPARATOR ', ') AS
CommonPreferences
FROM Guest g
LEFT JOIN Reservation r ON g.GuestID = r.GuestID
WHERE r.Status NOT IN ('Cancelled')
GROUP BY g.Nationality
ORDER BY TotalGuests DESC;
```

Sample Output:

Nationality	TotalGuests	TotalBookings	TotalSpent	AverageSpent
USA	2	3	$2,480.00	$826.67
Spain	1	2	$1,580.00	$790.00
Tanzania	2	2	$520.00	$260.00
UK	1	1	$1,400.00	$1,400.00
China	1	1	$1,080.00	$1,080.00

Report 5: Staff Performance Report

Purpose: Evaluates staff performance based on check-in/out handling.

```
Report 5: Staff Performance - Check-in/out Activity
SELECT
    s.StaffID,
    CONCAT(s.FirstName, ' ', s.LastName) AS StaffName,
    s.Position,
    COUNT(ci.CheckID) AS GuestsHandled,
    SUM(ci.FinalAmount) AS TotalRevenueHandled,
    AVG(ci.FinalAmount) AS AverageTransactionValue,
    MIN(ci.CheckInTime) AS FirstCheckIn,
    MAX(ci.CheckInTime) AS LastCheckIn
FROM Staff s
LEFT JOIN CheckInOut ci ON s.StaffID = ci.StaffID
GROUP BY s.StaffID
ORDER BY GuestsHandled DESC;
```

Sample Output:

Staff Name	Position	Guests Handled	Total Revenue Handled	Avg Transaction Value
James Mushi	Front Desk Supervisor	3	$3,080.00	$1,026.67
Grace Kiwia	Senior Receptionist	1	$900.00	$900.00
Sarah Daudi	Accountant	1	$1,400.00	$1,400.00

Report 6: Service Revenue Report

Purpose: Shows which services generate the most revenue.

```
Report 6: Service Revenue Analysis
SELECT
    s.ServiceName,
    s.Category,
    COUNT(so.ServiceOrderID) AS TimesOrdered,
    SUM(so.Quantity) AS TotalQuantity,
    SUM(so.TotalPrice) AS TotalRevenue,
    AVG(so.TotalPrice) AS AverageOrderValue,
```

```
    ROUND((SUM(so.TotalPrice) / (SELECT SUM(TotalPrice) FROM
ServiceOrder
        WHERE Status = 'Delivered')) * 100, 1) AS
RevenuePercentage
FROM Service s
LEFT JOIN ServiceOrder so ON s.ServiceID = so.ServiceID
WHERE so.Status = 'Delivered' OR so. Status IS NULL
GROUP BY s.ServiceID
ORDER BY TotalRevenue DESC;
```

Sample Output:

ServiceName	Category	TimesOrdered	TotalRevenue	RevenuePercentage
Spa Massage	Wellness	4	$300.00	34.5%
Room Service	Food	5	$75.00	8.6%
Breakfast Buffet	Food	3	$150.00	17.2%
Laundry Service	Cleaning	1	$20.00	2.3%
Excursion Tour	Entertainment	1	$80.00	9.2%

3. Strategic Analysis

3.1 How These Reports Help Management

Financial Management

Report	How It Helps Management
Monthly Revenue Report	Tracks income trends, identifies peak revenue periods, and helps with budgeting and forecasting.
Payment Method Analysis	Helps decide which payment systems to invest in and negotiate better rates with payment providers.

Room Management

Report	How It Helps Management
Room Occupancy Report	Identifies under performing room types and helps with pricing strategies.
Most Popular Rooms Report	Shows which rooms to promote and which need renovation or repricing.

Guest Management

Report	How It Helps Management
Guest Demographics Report	Identifies target markets and helps with marketing campaigns.
Guest Loyalty Report	Helps create loyalty programs and VIP treatment for repeat guests.

Service Management

Report	How It Helps Management
Service Revenue Report	Identifies profitable services and helps decide which services to promote or discontinue.
Daily Operations Summary	Provides real-time operational awareness for daily decision-making.

Staff Management

Report	How It Helps Management
Staff Performance Report	Identifies high-performing staff and helps with training needs and performance bonuses.

3.2 How the Database Supports Organizational Strategy

Strategic Area	How the Database Supports It
Revenue Optimization	Tracks all income sources and identifies the highest revenue generators.
Cost Reduction	Identifies inefficiencies, such as empty rooms and low-demand services.

Strategic Area	How the Database Supports It
Guest Satisfaction	Stores guest preferences, enabling personalized service.
Competitive Advantage	Data-driven decisions help the hotel outperform competitors.
Marketing Effectiveness	Guest demographics guide targeted campaigns.
Operational Efficiency	Real-time data speeds up processes such as check-in and reservations.
Scalability	The database grows with the hotel and supports future expansion.

3.3 How Managers Can Use Information for Decision-Making

1. Strategic Planning Decisions

Data Source	Decision
Revenue reports	Set annual revenue targets and budgets.
Occupancy trends	Plan expansions or renovations during low seasons.
Guest demographics	Enter new markets or launch targeted campaigns.

2. Tactical Decisions

Data Source	Decision
Room occupancy	Adjust daily room prices (dynamic pricing).
Staff performance	Schedule staff based on occupancy projections.
Service revenue	Run promotions on underperforming services.

3. Operational Decisions

Data Source	Decision
Daily operations	Allocate housekeeping based on check-in/out volume.
Payment methods	Offer discounts for preferred payment methods.
Service orders	Ensure popular services are always available.

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